Lab 6: Code and Branch Coverage Analysis

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Before

Run tests:

```
PS C:\Users\roopa\Desktop\clgcoding\SE-Labs-PES1UG24CS386\LAB_6> python -m pytest -v
========================================== test session starts ==========================================
platform win32 -- Python 3.12.4, pytest-8.3.4, pluggy-1.5.0 -- C:\Users\roopa\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\roopa\Desktop\clgcoding\SE-Labs-PES1UG24CS386\LAB_6
plugins: anyio-4.12.0
collected 2 items

test_hotel_booking.py::test_standard_two_nights PASSED                                          [ 50%]
test_hotel_booking.py::test_full_refund PASSED                                                  [100%]

========================================== 2 passed in 0.11s ==========================================
```

Line coverage:

```
PS C:\Users\roopa\Desktop\clgcoding\SE-Labs-PES1UG24CS386\LAB_6> coverage report -m
Name                    Stmts   Miss  Cover   Missing
-----------------------------------------------------
hotel_booking.py           32     18    44%   6, 8, 11, 14, 16, 23-29, 33-39
test_hotel_booking.py       6      0   100%
-----------------------------------------------------
TOTAL                      38     18    53%
```

Branch coverage:

```
PS C:\Users\roopa\Desktop\clgcoding\SE-Labs-PES1UG24CS386\LAB_6> coverage run --branch -m pytest
========================================== test session starts ==========================================
platform win32 -- Python 3.12.4, pytest-8.3.4, pluggy-1.5.0
rootdir: C:\Users\roopa\Desktop\clgcoding\SE-Labs-PES1UG24CS386\LAB_6
plugins: anyio-4.12.0
collected 2 items

test_hotel_booking.py ..                                                                        [100%]

========================================== 2 passed in 0.16s ==========================================
PS C:\Users\roopa\Desktop\clgcoding\SE-Labs-PES1UG24CS386\LAB_6> coverage report -m
Name                    Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------
hotel_booking.py           32     18     22      6    37%   6, 8, 11, 14, 16, 23-29, 33-39
test_hotel_booking.py       6      0      0      0   100%
-----------------------------------------------------------------
TOTAL                      38     18     22      6    43%
```

After

Code coverage:

```
PS C:\Users\roopa\Desktop\clgcoding\SE-Labs-PES1UG24CS386\LAB_6> coverage run --source=hotel_booking -m pytest CodeCoverageTest_hotel_booking_application.py
============================= test session starts =============================
platform win32 -- Python 3.12.4, pytest-8.3.4, pluggy-1.5.0
rootdir: C:\Users\roopa\Desktop\clgcoding\SE-Labs-PES1UG24CS386\LAB_6
plugins: anyio-4.12.0
collected 19 items

CodeCoverageTest_hotel_booking_application.py ...................        [100%]

============================= 19 passed in 0.22s ==============================
```

```
PS C:\Users\roopa\Desktop\clgcoding\SE-Labs-PES1UG24CS386\LAB_6> coverage report -m
Name                Stmts   Miss  Cover   Missing
-------------------------------------------------
hotel_booking.py       32      0   100%
-------------------------------------------------
TOTAL                  32      0   100%
```

Branch Coverage:

```
PS C:\Users\roopa\Desktop\clgcoding\SE-Labs-PES1UG24CS386\LAB_6> coverage run --branch --source=hotel_booking -m pytest BranchCoverageTest_hotel_booking_appli
cation.py
============================= test session starts =============================
platform win32 -- Python 3.12.4, pytest-8.3.4, pluggy-1.5.0
rootdir: C:\Users\roopa\Desktop\clgcoding\SE-Labs-PES1UG24CS386\LAB_6
plugins: anyio-4.12.0
collected 24 items

BranchCoverageTest_hotel_booking_application.py ........................        [100%]

============================= 24 passed in 0.24s ==============================
PS C:\Users\roopa\Desktop\clgcoding\SE-Labs-PES1UG24CS386\LAB_6> coverage report -m
Name              Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------
hotel_booking.py      32      0     22      0   100%
-----------------------------------------------------------
TOTAL                 32      0     22      0   100%
```